AS · Cambridge (CIE) · Physics

22 mins 22 questions

Multiple Choice Questions

SI Units

SI Units / Homogeneity of Physical Equations & Powers of Ten

Easy (8 questions)	/8
Medium (7 questions)	/7
Hard (7 questions)	/7
Total Marks	**/22**

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Easy Questions

1 Which of the following is an SI base unit?

A. Current

B. Gram

C. Kelvin

D. Volt

(1 mark)

2 Five energies are listed below.

5kJ

5mJ

5MJ

5nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

A. $5\,\text{kJ} \rightarrow 5\,\text{mJ} \rightarrow 5\,\text{MJ} \rightarrow 5\,\text{nJ}$

B. $5\,\text{nJ} \rightarrow 5\,\text{kJ} \rightarrow 5\,\text{MJ} \rightarrow 5\,\text{mJ}$

C. $5\,\text{nJ} \rightarrow 5\,\text{mJ} \rightarrow 5\,\text{kJ} \rightarrow 5\,\text{MJ}$

D. $5\,\text{mJ} \rightarrow 5\,\text{nJ} \rightarrow 5\,\text{kJ} \rightarrow 5\,\text{MJ}$

(1 mark)

3 Which product-pair of metric prefixes has the greatest magnitude?

A. pico × mega

B. nano × kilo

C. micro × giga

D. milli × tera

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(1 mark)

4 Which statement includes a correct unit?

A. Energy = 7.8 N s

B. Force = 3.8 N s

C. Momentum = 6.2 N s

D. Torque = 4.7 N s

(1 mark)

5 Which quantity can be measured in electronvolts (eV)?

A. Electric charge

B. Electric potential

C. Energy

D. Power

(1 mark)

6 Which of the following is not a SI base unit?

A. kilogram

B. metre

C. second

D. volt

(1 mark)

7 Which product pair of metric prefixes has the smallest magnitude?

A. nano × mega

B. tera × pico

C. centi × kilo

D. micro × giga

(1 mark)

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8 Which unit is **not** an SI base unit?

A. A

B. K

C. mol

D. N

(1 mark)

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Medium Questions

1 The average kinetic energy E of a gas molecule is given by the equation

$$E = \frac{3}{2}kT$$

where T is the thermodynamic (kelvin) temperature.

What are the SI base units of k?

A. $\text{kg}^{-1}\ \text{m}^{-1}\ \text{s}^{2}\ \text{K}$

B. $\text{kg}^{-1}\ \text{m}^{-2}\ \text{s}^{2}\ \text{K}$

C. $\text{kg m s}^{-2}\ \text{K}^{-1}$

D. $\text{kg m}^{2}\ \text{s}^{-2}\ \text{K}^{-1}$

(1 mark)

2 The units of all physical quantities can be expressed in terms of SI base units.

Which pair contains quantities with the same base units?

A. Force and momentum

B. Pressure and Young modulus

C. Power and kinetic energy

D. Mass and weight

(1 mark)

3 The table contains some quantities, together with their symbols and units.

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Quantity	Symbol	Unit
Gravitational field strength	g	N kg^{-1}
Density of liquid	ρ	kg m^{-3}
Vertical height	h	m
Volume of part of liquid	V	m^3

Which expression has the units of energy?

A. $\rho g^2 h$

B. $\frac{\rho h V}{g}$

C. $\frac{\rho g}{h V}$

D. $g \rho h V$

(1 mark)

4 What is the ratio $\frac{10^{-3}\,\text{THz}}{10^{3}\,\text{kHz}}$?

A. 10^{-9}

B. 10^{-6}

C. 10^{0}

D. 10^{3}

(1 mark)

5 Which of the following correctly expresses the volt in terms of SI base units?

A. AΩ

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B. WA^{-1}

C. $\text{kg m}^2\ \text{s}^{-1}\ \text{A}^{-1}$

D. $\text{kg m}^2\ \text{s}^{-3}\ \text{A}^{-1}$

(1 mark)

6 The equation relating pressure and density is $P = \rho gh$

How can both sides of this equation be written in terms of base units?

A. $[\text{N m}^{-1}] = [\text{kg m}^{-3}][\text{m s}^{-1}][\text{m}]$

B. $[\text{N m}^{-2}] = [\text{kg m}^{-2}][\text{m s}^{-2}][\text{m}]$

C. $[\text{kg m}^{-1}\ \text{s}^{-2}] = [\text{kg m}^{-3}][\text{m s}^{-2}][\text{m}]$

D. $[\text{kg m}^{-1}\ \text{s}^{-1}] = [\text{kg m}^{-1}][\text{m s}^{-2}][\text{m}]$

(1 mark)

7 What is the ratio $\frac{10^2\ \text{THz}}{10^{-4}\ \text{pHz}}$?

A. 10^{-30}

B. 10^{-2}

C. 10^{2}

D. 10^{30}

(1 mark)

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Hard Questions

1 The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation

$$v = k\left(\frac{\Delta P}{\rho}\right)^n$$

where k is a constant that has no units.

What is the value of n?

A. $\frac{1}{2}$

B. 1

C. $\frac{3}{2}$

D. 2

(1 mark)

2 The frictional force F on a sphere falling through a fluid is given by the formula

$$F = 6\pi r\eta v$$

where r is the radius of the sphere, η is a constant relating to the fluid and v is the velocity of the sphere.

What are the units of η ?

A. kg m s^{-1}

B. $\text{kg m}^{-1}\text{ s}^{-1}$

C. kg m s^{-3}

D. $\text{kg m}^{3}\text{ s}^{-3}$

(1 mark)

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3 When a constant braking force is applied to a vehicle moving at speed v, the distance d moved by the vehicle as it comes to rest is given by the expression

$$d = kv^2$$

where k is a constant.

When d is measured in metres and v is measured in metres per second, the constant has a value of k_1.

What is the value of the constant k when the distance is measured in metres, and the speed is measured in kilometres per hour?

A. $0.077k_1$

B. $0.278k_1$

C. $3.60k_1$

D. $13.0k_1$

(1 mark)

4 The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n?

A. 1

B. 2

C. 3

D. 4

(1 mark)

5 Which formula could be correct for the speed v of ocean waves in terms of the density ρ of sea-water, the acceleration of free fall g, the height h of the wave and the wavelength λ?

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A. $v = \sqrt{\rho g h}$

B. $v = \sqrt{\frac{g}{h}}$

C. $v = \sqrt{g\lambda}$

D. $v = \sqrt{\frac{g}{\rho}}$

(1 mark)

6 The equation for the terminal velocity of a sphere of radius *r* falling under acceleration due to gravity, *g*, through a liquid with density ρ and viscosity η, measured in kg m^{-1} s^{-1} is:

$$v_t = \frac{2}{9}\frac{gr^n\rho}{\eta}$$

What is the value of *n*?

A. 0.5

B. 1

C. 2

D. 3

(1 mark)

7 The current I through a semiconductor diode is given by:

$$I = I_0 e^{\frac{-qV}{nT}}$$

where I_0 is the initial current, q is the charge of an electron, V is the applied potential difference, T is the temperature, and n is a constant.

What is the unit of n when expressed in SI base units?

A. kg m^2 s^{-2} K^{-1}

B. kg m^2 s^{-4} K^{-1}

C. kg^{-1} m^{-2} s^2 K

D. kg^{-1} m^{-2} s^4 K

(1 mark)

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